Contents

Chapter 1

Brownian Motion in Soft Confinement

1.1 Introduction

In many biological systems, one must not only take into account Brownian behaviours, but also carefully evaluate the effect of the system's or parts of the system's deformability on its dynamics. Indeed, if we take the example of a lipid bilayer, which is a soft and quasi-two-dimensional fluid, the thermal energy $k_{\mathrm{B}}\Theta$ of the system is sufficient to cause long-wavelength bending fluctuations of the cell membrane, thickness fluctuations, lipid movement within the plane itself, and so forth. Naturally, a neighbouring fluid's motion is then affected by these shape modifications, which can in turn affect the dynamics of a body evolving in that same bath. This coupling is not trivial. I describe in this chapter what has already been developed on that matter, starting with the more well-established case of deterministic soft lubrication. I then present an overview of the stochastic case, where bodies or confinements are fluctuating and deformable. The remainder of the chapter is articulated around the presentation, development and numerical verifications of a simple and elementary model we developed during my doctoral contract to assess this same problematique of soft Brownian effects.

1.1.1 Deterministic soft lubrication

1.1.2 Stochastic soft lubrication

The aim of this chapter is to probe at these modifications to colloid's behaviour.

Because of the level of complexity of such a study, I present in this chapter a most simple model, with elementary ingredients, which can still qualitatively capture all the effects we are interested in. Such model is detailed in the next section. In the following sections, I develop the numerical model used to simulate this toy model. I then present the theoretical aspects of the model, developed by David S. Dean. Finally, I show how numerics and theory compare.

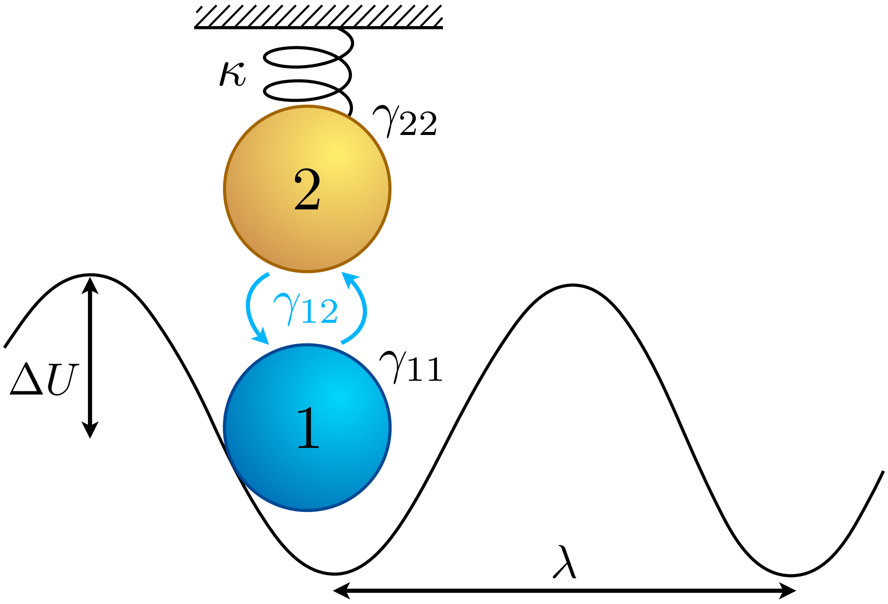

Figure 1.1: Representation of the system. A Brownian colloid of interest (blue) is trapped in a periodic potential of amplitude ΔU and spatial period λ. An elastic membrane is modelled as a second Brownian colloid (orange) subject to a harmonic potential of inverse stiffness of compliance κ. This membrane is coupled to the colloid of interest through an indirect hydrodynamic coupling represented by the coefficient γ_{12}, and both the colloid and the membrane feel their own self-friction γ_{11} and $\gamma 22$, respectively.

1.2 Toy Model for soft Brownian Motion

1.2.1 Overview of the Toy Model

From here on, we study a purely one-dimensional case of a Brownian colloid coupled to an elastic and fluctuating membrane (Fig. 1.1). The Brownian colloid is represented by a first degree of freedom, q_1, and is submitted to a periodic trapping potential $\phi(q_1)$ of spatial period λ and amplitude ΔU. We couple to this colloid the toy model version of the membrane. We represent the membrane with a colloid that is submitted in a harmonic potential of energy $\frac{q_2^2}{2\kappa}$, where κ is the compliance (or inverse stiffness). It represents the elasticity of the membrane. This second colloid is tracked with the second degree of freedom, q_2.

Both the Brownian colloid and the membrane feel their own self-friction, respectively γ_{11} and $\gamma_2 2$. Furthermore, they interact through and indirect, hydrodynamic-like interaction γ_{12}. This interaction represents the effect of the fluid that is displaced by the motion of the colloid on the membrane, and *vice-versa*. All together, these terms form the friction matrix $\mathbf{\Gamma}$

$$\mathbf{\Gamma} = \begin{pmatrix} \gamma_{11} & \gamma_{12} \\ \gamma_{12} & \gamma_{22} \end{pmatrix} \tag{1.1}$$

which can be inverted to give the mobility matrix of the system, as $\mathbf{M} = \mathbf{\Gamma}^{-1}$. Both matrices must be positive definite, and, because their diagonal terms account for the self-friction and self-mobility respectively, $\mu_{11} > 0$ and $\mu_{22} > 0$.

We can write the total energy as

$$H(q_1, q_2) = \phi(q_1) + \frac{q_2^2}{2\kappa} \ , \tag{1.2}$$

and we notice the absence of Hamiltonian interactions in the system. In that case, we work in the overdamped limit, and we write the coupled Langevin equations describing the evolutions of q_1 and q_2 as

$$\gamma_{11}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \gamma_{12}\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\phi'(q_1) + \eta_1(t) \ , \tag{1.3}$$

$$\gamma_{12}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \gamma_{22}\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\frac{q_2}{\kappa} + \eta_2(t) \ , \tag{1.4}$$

where η_i and η_j (with $j \in \{1, 2\}$) are Gaussian white noises of zero mean and correlation functions $\langle \eta_i(t)\eta_j(t')\rangle = 2k_\mathrm{B}T\delta(t - t')\gamma_{ij}$.

$$P_\mathrm{eq} = \frac{\exp[-\beta H(q_1, q_2)]}{Z} \ , \tag{1.5}$$

1.3 Theoretical developments

The objective of the following developments is to compute the modification to the long-time dynamics of the Brownian colloid diffusing withing the sinusoidal trap. It is worthwhile to note that our published work presents several derivations to do so, and the method I present here is not strictly rigorous. I invite the interested reader to consult the supplementary material of our publication, which can be found in the appendix of this manuscript.

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_{11}\phi'(q_1) - \frac{\mu_{12}}{\kappa}q_2 \ , \tag{1.6}$$

$$\frac{\mathrm{d}q_2}{\mathrm{d}t} = -\mu_{12}\phi'(q_1) - \frac{\mu_{22}}{\kappa}q_2 \tag{1.7}$$

$$P_\mathrm{meq}(q_1) = \frac{\exp[-\beta\phi(q_1)}{Z_-} \ , \tag{1.8}$$

$$Z_\pm = \int_0^\lambda \mathrm{d}q \exp[\pm\beta\phi(q)] \ , \tag{1.9}$$

$$q_2 = -\frac{\kappa\mu_{12}}{\mu_{22}}\left(1+\frac{\kappa}{\mu_2 2}\frac{\mathrm{d}}{\mathrm{d}t}\right)^{-1}\phi'(q_1) \tag{1.10}$$

$$q_2 \simeq -\frac{\kappa\mu_{12}}{\mu_{22}}\left[\phi'(q_1)-\frac{\kappa}{\mu_2 2}\phi''(q_1)\frac{\mathrm{d}q_1}{\mathrm{d}t}+O(\tau_2^2)\right] \tag{1.11}$$

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\left(\mu_{11}-\frac{\mu_{12}^2}{\mu_{22}}\right)\phi'(q_1(t))-\frac{\kappa\mu_{12}^2}{\mu_{22}^2}\phi''(q1(t))\frac{\mathrm{d}q_1}{\mathrm{d}t}\ , \tag{1.12}$$

$$\frac{\mathrm{d}q_1}{\mathrm{d}t} = -\mu_{\mathrm{e}}(q_1)\phi'(q_1)\ , \tag{1.13}$$

$$\mu_{\mathrm{e}} = \frac{\mu_{11}\mu_{22}-\mu_{12}^2}{\mu_{22}\left(1+\frac{\kappa\phi''(q_1)\mu_{12}^2}{\mu_{22}^2}\right)} \tag{1.14}$$

$$\frac{\partial P_1(q_1,t)}{\partial t} = \frac{\partial}{\partial q_1}\mu_{\mathrm{e}}(q_1)\left[k_{\mathrm{B}}\Theta\frac{\partial P_1(q_1,t)}{\partial q_1}+\phi'(q_1)P_1(q_1,t)\right] \tag{1.15}$$

$$D^* = \frac{\lambda^2}{\int_0^\lambda \mathrm{d}q\frac{\exp[\beta\phi(q)]}{D_{\mathrm{e}}}\int_0^\lambda \mathrm{d}q\exp[-\beta\phi(q)]} \tag{1.16}$$

$$D^* = \frac{k_{\mathrm{B}}\Theta\lambda^2}{\gamma_{11}\int_0^\lambda \mathrm{d}q\exp[\beta\phi(q)]\left(1-\frac{\kappa\gamma_{12}^2\phi'^2(q)}{k_{\mathrm{B}}\Theta\gamma_{11}^2}\right)\int_0^\lambda \mathrm{d}q\exp[-\beta\phi(q)]} \tag{1.17}$$

here.

$$D^* = \frac{k_{\mathrm{B}}\Theta}{\gamma_{11}\int_0^1 \mathrm{d}p\exp[\beta\Delta U\psi(p)]\left(1-\frac{\kappa\gamma_{12}^2\Delta U^2\psi'^2(p)}{k_{\mathrm{B}}\Theta\gamma_{11}^2}\right)\int_0^1 \mathrm{d}p\exp[-\beta\Delta U\psi(p)]} \tag{1.18}$$

1.4 Numerical modelling

1.4.1 Discretisation

My main mission in this project is to simulate the above Eqs. 1.3 and 1.4 efficiently. To do so, I discretise these equations using a simple forward Euler scheme (which was introduced in the beginning of this thesis). The discretised equations read

$$q_1(t+\Delta t)-q_1(t) = -\mu_{11}\phi'(q_1(t))\Delta t-\frac{\mu_{12}}{\kappa}q_2(t)\Delta t+\xi_1\sqrt{2k_{\mathrm{B}}T\Delta t}\ , \tag{1.19}$$

$$q_2(t+\Delta t)-q_2(t) = -\frac{\mu_{22}}{\kappa}q_2(t)\Delta t-\mu_{12}\phi'(q_1(t))\Delta t+\xi_2\sqrt{2k_{\mathrm{B}}T\Delta t} \tag{1.20}$$

where ξ_i is a zero-mean Gaussian random variable, as

$$\langle\xi_i\xi_j\rangle = \mu_{ij}\ . \tag{1.21}$$

We choose to work in terms of mobility instead of friction matrix elements for convenience in the upcoming developments. They are, however, strictly equivalent.

Furthermore, in the rest of this study, we choose $\phi(q_1)$ to be sinusoidal, as $\phi(q_1) = \Delta U \sin\left(\frac{2\pi q_1}{\lambda}\right)$, with ΔU the amplitude of the sinusoidal trap.

1.4.2 Time step selection

In any discretisation process, it is essential to choose the appropriate time step. This time step must be small enough to avoid a blowup from error propagation (again, see the beginning of this thesis for more details on how the Euler method propagates errors), but big enough to remain advantageous. An arbitrarily small timestep may prohibit the probing of very long time dynamics due to computation time.

To select a correct time step for a simulation, I typically first identify all the smaller time scales in the system I want to simulate. In the present study, I identify two time scales : the local relaxation time within a local minimum of the sinusoidal trap, and the relaxation time of the harmonic trap.

Near a local minimum, we may approximate the sinusoidal trap as harmonic. In that case,

$$\phi(q_1) = \frac{1}{2} k_{\text{eff}} q_1^2 \ , \tag{1.22}$$

where k_{eff} is an effective stiffness, which we identify as the local curvature of the potential as

$$\begin{aligned} k_{\text{eff}} &= |\phi''(0)| & (1.23) \\ &= \Delta U \left(\frac{2\pi}{\lambda}\right)^2 \ , & (1.24) \end{aligned}$$

and we can finally identify this first timescale as

$$\tau_1 = (\mu_{11} k_{\text{eff}})^{-1} = \frac{\lambda^2}{4\pi^2 \Delta U \mu_{11}} \ . \tag{1.25}$$

The harmonic trap is much simpler, as it requires no approximations, and we find its characteristic timescale

$$\tau_2 = \frac{\kappa}{\mu_{22}} \ . \tag{1.26}$$

Then, I choose a time step which satisfies

$$\Delta t = \frac{1}{100} \max \tau_1, \tau_2 \ . \tag{1.27}$$

1.5 Results

1.6 Conclusion and perspectives on the chapter